

COMPUTER SCIENCE**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 12 printed pages.
4. It is divided into two parts: Part I and Part II.
5. It has eleven questions in all.
6. Part I is compulsory and has two questions.
7. While attempting Multiple Choice Questions in Part I, you are required to write only ONE option as the answer.
8. Part II is divided into three sections: A, B and C.
9. Each section in Part II has three questions. Any two questions have to be attempted from each section.
10. The intended marks for questions are given in brackets [].

Instruction to Supervising Examiner

1. Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

PART I – 20 MARKS*Answer all questions.**While answering questions in this Part, briefly indicate your working and reasoning, wherever required.***Question 1**

- (i) According to the Principle of Duality, the Boolean equation $X + X' \cdot Y + 0 = X + Y$ will be equivalent to: (*Application*) [1]
- (a) $X \cdot (X' + Y) \cdot 1 = X \cdot Y$
 - (b) $X' \cdot (X + Y') \cdot 1 = X' \cdot Y'$
 - (c) $X \cdot (X' + Y) \cdot 0 = X \cdot Y$
 - (d) $X + (X' \cdot Y) + 1 = X + Y$
- (ii) The complement of the Boolean expression $(P' + Q) \cdot (Q' + R')$ is: (*Application*) [1]
- (a) $P \cdot Q' + Q \cdot R$
 - (b) $P' \cdot Q + Q' \cdot R'$
 - (c) $(P + Q') \cdot (Q + R)$
 - (d) $P \cdot Q' \cdot Q \cdot R$

- (iii) The Boolean equations $A \cdot A' = 0$ and $A + A' = 1$ correspond to: (*Recall*) [1]
- (a) Involution law
 - (b) Complement law
 - (c) Idempotent law
 - (d) Law of identity
- (iv) The reduced expression of $F(A, B) = \Sigma(1, 2)$ is: (*Application*) [1]
- (a) $A \odot B$
 - (b) $A \oplus B$
 - (c) $A + B$
 - (d) $A' \cdot B'$
- (v) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]
- Assertion:** After the statements `StringBuffer sb = new StringBuffer("GOOD"); sb.append("BYE");` the object referred to by `sb` holds "GOODBYE".
- Reason:** StringBuffer objects are mutable.
- (*Analysis*)
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Both Assertion and Reason are false.
- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]
- Assertion:** If the user types `hello` when `sc.nextInt( )` is executed (`sc` being a Scanner object), an `InputMismatchException` is thrown.
- Reason:** `InputMismatchException` is a checked exception, so every method that calls `nextInt( )` must declare it with a `throws` clause.
- (*Analysis*)
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Both Assertion and Reason are false.

(vii) Consider the following class:

[1]

```
class Counter
{
    static int total;
    int id;
    static void report()
    {
        ...
    }
}
```

Which of the following statements are valid for the method `report()`?

- I. It can use the data member `total` directly.
- II. It can use `this.id` to refer to the id of the current object.
- III. It can be invoked as `Counter.report()` without creating any object.

(Application)

- (a) I and II
- (b) II and III
- (c) I and III
- (d) Only III

(viii) The complexity of the following code segment is:

[1]

```
for (int i = 0; i < n; i++)
    for (int j = 0; j < 5; j++)
        s = s + a[i] * j;
```

(Analysis)

- (a) $O(1)$
- (b) $O(n)$
- (c) $O(n^2)$
- (d) $O(5n^2)$

(ix) Write the inverse of the statement: "If the traffic signal is red, then the vehicles stop."

[1]

(Application)

(x) Why are NAND and NOR gates called universal gates? (Understanding)

[1]

Question 2

(i) Convert the following infix notation to postfix form:

[2]

$(A + B \wedge C \wedge D) * E - F / G$ (Application)

(ii) A two-dimensional array `N[15][10]` (15 rows and 10 columns) is stored in the memory row major wise, each element requiring 8 bytes. If the element `N[2][1]` is stored at the address 4200, find the address of `N[9][6]`. (Application)

[2]

- (iii) The following function test() is a part of some class. Assume n is a positive integer. Answer the questions given below along with dry run / working.

```
void test(int n)
{
    if (n > 0)
    {
        System.out.print(n + " ");
        test(n - 2);
        System.out.print(n * 10 + " ");
    }
}
```

- (a) What will be printed when test(5) is called? (*Analysis*) [2]
(b) What is the function test() performing apart from recursion? (*Analysis*) [1]
- (iv) The class Savings below inherits the class Account and displays the interest along with the balance. There are some places in the code marked by ?1?, ?2?, ?3? which may be replaced by a keyword / statement so that the classes work properly.

```
class Account
{
    double bal;
    Account(double b)
    {
        bal = b;
    }
    void display()
    {
        System.out.println("Balance : " + bal);
    }
}
class Savings ?1? Account
{
    double rate;
    Savings(double b, double r)
    {
        ?2?;
        rate = r;
    }
    void display()
    {
        ?3?;
        System.out.println("Interest : " + bal * rate / 100);
    }
}
```

- (a) What is the keyword at ?1? (*Analysis*) [1]
(b) What is the statement at ?2? (*Analysis*) [1]
(c) What is the statement at ?3? (*Analysis*) [1]

PART II – 50 MARKS

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION A

Answer any two questions.

Question 3

- (i) A school admits a student to its Robotics Club (J) if any one of the following conditions is satisfied: [5]

- The student has opted for Computer Science and has the consent of the parents, but is not a member of any other club.

OR

- The student has scored above 80% in Mathematics and has opted for Computer Science.

OR

- The student has scored above 80% in Mathematics and has the consent of the parents, but is not a member of any other club.

The inputs are:

INPUTS

M The student has scored above 80% in Mathematics.

C The student has opted for Computer Science.

P The student has the consent of the parents.

E The student is a member of some other club.

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: J – denotes that the student is admitted [1 indicates yes and 0 indicates no in all cases].

Draw the truth table for the inputs and outputs given above and write the SOP expression for $J(M, C, P, E)$.

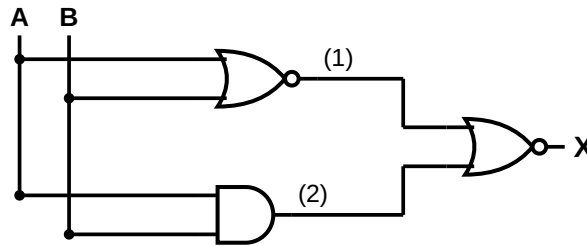
(Application)

- (ii) Reduce the above expression $J(M, C, P, E)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. (Create) [5]

Question 4

- (i)
- (a) Reduce the Boolean function $F(P, Q, R, S) = P'Q'R'S' + P'Q'R'S + P'Q'RS' + P'Q'RS + P'QR'S + PQ'R'S' + PQ'R'S + PQ'RS' + PQ'RS + PQR'S$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). (Application) [4]
- (b) Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. (Create) [1]

- (ii) From the logic diagram given below, derive the Boolean expressions at (1), (2) and X, reduce X and name the single gate it is equivalent to. (*Application*) [3]



- (iii) Simplify the following expression using the laws of Boolean algebra, naming the law used at each step: $A \cdot B + A \cdot B' \cdot C + A' \cdot C$ (*Application*) [2]

Question 5

- (i)
- How many select lines does an 8 : 1 multiplexer have? (*Recall*) [1]
 - Write the output expression of an 8 : 1 multiplexer with select lines A, B, C and data inputs I_0 to I_7 . Show, with a block diagram, how it can be used to realise the function $F(A, B, C) = \Sigma(1, 2, 4, 7)$ by connecting each data input to 0 or 1. (*Application*) [3]
 - Identify the function F realised above. (*Analysis*) [1]
- (ii) Draw the logic circuit of a half adder using NAND gates only. Show the expression at the output of each gate. (*Create*) [3]
- (iii) Convert the cardinal form $F(P, Q, R) = \pi(0, 2, 5)$ into its canonical POS form, and also write the canonical SOP form of the same function. (*Application*) [2]

SECTION B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem.

This can be achieved by using mnemonic names and comments in the program.

(Flowcharts and Algorithms are not required.)

The programs must be written in Java.

Question 6

A **Happy number** is one which eventually reaches 1 when it is replaced, again and again, by the sum of the squares of its digits. If the process reaches 4 instead, it then repeats the cycle 4, 16, 37, 58, 89, 145, 42, 20, 4 for ever and the number is not happy. [10]

Example 1: $28 \rightarrow 4 + 64 = 68 \rightarrow 36 + 64 = 100 \rightarrow 1 + 0 + 0 = 1$
so 28 is a HAPPY number

Example 2: $20 \rightarrow 4 + 0 = 4$, so 20 is NOT a happy number

Design a class Happy to check whether a given positive number is a happy number. Some of the members of the class are given below:

Class name : **Happy**

Data members/instance variables:

n : to store the number

Methods/Member functions:

Happy() : default constructor to initialise the data member with a legal initial value

void getnum(int nn) : to assign the parameter value nn to the number n

int sum_sq_digits(int x) : returns the sum of the squares of the digits of x using the **recursive technique**

void ishappy() : checks whether n is a happy number by invoking sum_sq_digits() repeatedly, and displays an appropriate message

Specify the class Happy giving details of the constructor(), void getnum(int), int sum_sq_digits(int) and void ishappy(). Define a main() function to create an object and call the functions accordingly to enable the task.

(Create)

Question 7

A class Angle stores an angle in degrees, minutes and seconds in an array of three integers (60 seconds = 1 minute, 60 minutes = 1 degree). Two angles are added and their sum is returned as an object. [10]

Example: First angle : 45 deg 50 min 40 sec
Second angle : 30 deg 25 min 35 sec
Sum : 76 deg 16 min 15 sec

The details of the members of the class are given below:

Class name : **Angle**

Data members/instance variables:

arr[] : integer array of 3 elements to store the degrees, minutes and seconds

Methods/Member functions:

Angle() : default constructor to initialise the array elements to 0

void readAngle() : to accept the degrees, minutes and seconds of an angle

Angle addAngle(Angle t) : to add the angle of the parameterised object t and the angle of the current object, store the sum in a local object and return it

void dispAngle() : to display the angle in degrees, minutes and seconds

Specify the class Angle giving details of the constructor(), void readAngle(), Angle addAngle(Angle) and void dispAngle(). Define a main() function to create objects and call the functions accordingly to enable the task.

(Create)

Question 8

A class VowelWord has been defined to count the words of a sentence which begin and end with a vowel, and to rearrange the sentence so that these words come first, in their original order, followed by the remaining words in their original order. The sentence is in UPPER CASE, its words are separated by a single blank space and it is terminated by '.', '?' or '!'.

[10]

Example:

```
INPUT  : ANAMIKA AND SUSAN ARE NEVER GOING TO QUARREL ANYMORE.  
OUTPUT : NUMBER OF WORDS BEGINNING AND ENDING WITH A VOWEL = 3  
        ANAMIKA ARE ANYMORE AND SUSAN NEVER GOING TO QUARREL
```

Some of the members of the class are given below:

Class name : **VowelWord**

Data members/instance variables:

sent	: to store the sentence
newSent	: to store the rearranged sentence
count	: to store the number of words beginning and ending with a vowel

Methods/Member functions:

VowelWord()	: default constructor to initialise the data members with legal initial values
void accept()	: to accept the sentence
boolean isVowelWord(String w)	: returns true if the word w begins and ends with a vowel, otherwise returns false
void arrange()	: counts the words that begin and end with a vowel and forms newSent as described above, invoking isVowelWord()
void display()	: displays the count and the rearranged sentence

Specify the class VowelWord giving details of the constructor(), void accept(), boolean isVowelWord(String), void arrange() and void display(). Define a main() function to create an object and call the functions accordingly to enable the task.

(Create)

SECTION C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise.

*This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms. The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.
(Flowcharts are not required.)*

Question 9

- (i) A network card stores incoming data packets (integers) in a buffer. Packets are added at the rear and removed from the front, and the positions freed at the front are reused, the rear end being connected to the front end in a circular arrangement. [4]

Define a class Buffer with the following details:

Class name : **Buffer**

Data members/instance variables:

pkt[] : array to hold the integer packets
cap : stores the maximum capacity of the array
front : to point to the index of the front
rear : to point to the index of the rear

Methods/Member functions:

Buffer(int max) : constructor to initialise cap = max, front = rear = 0 and to create the array
void insert(int v) : to add a packet at the rear if possible, otherwise display the message "BUFFER FULL"
int delete() : to remove and return the packet from the front; if the buffer is empty, return -9999
void show() : to display the packets from front to rear

Specify the class Buffer giving details of the functions void insert(int) and int delete().

Assume that the other functions have been defined.

The main() function and algorithm need NOT be written.

(Create)

- (ii) State one advantage of a circular queue over a linear queue. *(Understanding)* [1]

Question 10

An electricity board charges its domestic consumers as per the tariff given below. A fixed meter rent of Rs 50 is added to every bill.

[5]

Units consumed	Rate
First 100 units	Rs 5.00 per unit
Next 200 units (101 to 300)	Rs 6.50 per unit
Above 300 units	Rs 8.00 per unit

A super class Consumer has been defined to store the details of a consumer. Define a sub class Bill to calculate the amount to be paid. The details of the members of both the classes are given below:

Class name : **Consumer**

Data members/instance variables:

name : to store the name of the consumer
cno : to store the consumer number
units : to store the number of units consumed

Methods/Member functions:

Consumer(...) : parameterised constructor to assign values to the data members
void show() : to display the consumer details

Class name : **Bill**

Data members/instance variables:

amt : to store the amount to be paid (in decimals)

Methods/Member functions:

Bill(...) : parameterised constructor to assign values to the data members of both the classes
void calculate() : to calculate the amount as per the tariff, including the meter rent
void show() : to display the consumer details along with the amount to be paid

Assume that the super class Consumer has been defined. Using the concept of inheritance, specify the class Bill giving the details of the constructor(...), void calculate() and void show().

The super class, main function and algorithm need NOT be written.

(Create)

Question 11

- (i) A linked list is formed from the objects of the class Node. The class structure of the Node is given below: [2]

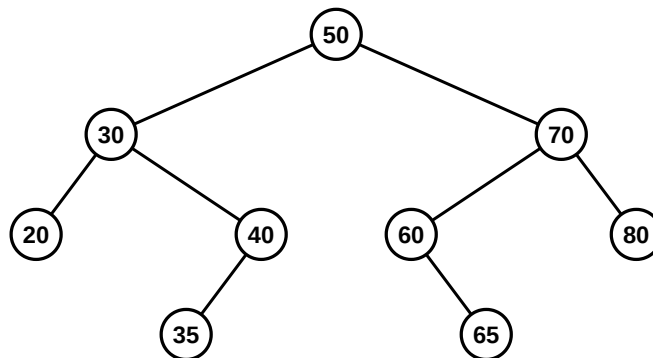
```
class Node
{
    int data;
    Node next;
}
```

Write an Algorithm OR a Method to reverse an existing linked list (the last node becomes the first) and return the reference to the first node of the reversed list. The method declaration is as follows:

Node reverse(Node start)

(Analysis)

- (ii) Answer the following questions from the diagram of a Binary Search Tree given below:



- (a) Write the postorder traversal of the tree. (Understanding) [1]
- (b) State the level of node 60 and the height of the tree (the root is at level 0). (Understanding) [1]
- (c) Name the internal nodes and the external nodes of the left subtree. (Understanding) [1]

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